

B. Sc. (Hons.) Semester-IV Examination, 2022-23
GEOLOGY

Paper: GLB-401: Palaeontology and Stratigraphy

Time: Three Hours

Full Marks: 70

Note: Answer five questions in all, selecting two each from Section-A and Section-B. Section-C is compulsory. All questions carry equal marks.

Section – A

Attempt any two of the following questions:

1. (a) Discuss the different physical and chemical conditions required for Fossilization. [7]
(b) With neat sketches, describe the structural morphology of a rhabdosome in graptolites. [7]
(c) With neat sketches, discuss the detailed skeletal morphology of Anthozoa. [7]
 2. With the help of neat, labeled sketches, discuss the dynamic skeletal morphology and geological distribution of the class Echinoidea. [14]
 3. Explain in detail the construction, divisions, and significance of the global Geological Time Scale. [14]
 4. Briefly explain the Cuddapah Supergroup of rocks with special emphasis on its lithostratigraphic classification, component lithology, and economic importance. [14]
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Section - B

Attempt any two of the following questions:

1. Write a brief geological note on any **two** of the following: [2 × 7 = 14]
 - (a) Processes of Mineralization in fossil preservation
 - (b) Spatial and geological distribution of trilobites
 - (c) Economic importance of the Dharwar Supergroup
 2. Write a brief explanatory note on any **two** of the following: [2 × 7 = 14]
 - (a) Morphology of the cephalon in trilobites (with neat sketches)
 - (b) Fundamental Principles of Stratigraphy
 - (c) Physical subdivisions of India (supported with a suitable sketch map)
 3. Describe in detail the stratigraphy of the Aravalli Craton of India, with special reference to its lithostratigraphic classification, field layout, and economic importance. [14]
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Section - C

Compulsory Section: Answer the following in 1 to 5 sentences each:

[7 × 2 = 14]

1. Explain the difference between a reworked fossil and a remain fossil.
 2. Define the term *Genal angle* as observed in trilobites.
 3. What is meant by an *Index fossil*?
 4. Define the structural role of a *Calyx* in Anthozoa.
 5. Write the name and location of a major barite deposit located within the Cuddapah Basin.
 6. What is a *gondite*?
 7. What is the precise geological age of the Rohtas Limestone?
 8. Write the name of a prominent gold deposit located within the Dharwar Craton.
 9. Write the names of two major peninsular rivers of India.
 10. Write the name of the oldest recognized rock unit within the Vindhyan Basin.
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External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>